SEMINAR 8

EXTREME CU LEGĂTURI

METODA MULTIPLICATORILOR LAGRANGE

1. Găsiți punctele de extrem local ale fct. $f:\mathbb{R}^2\to\mathbb{R}$, $f(x,y)=6-4x-3y$ cu legătura $x^2+y^2=1$.

SOL:

Fie $g:\mathbb{R}^2\to\mathbb{R}$, $g(x,y)=x^2+y^2-1$

Evid $f,g\in\mathcal{C}^2(\mathbb{R}^2)$, $\mathbb{R}^2$ - m.d.

(I) $\operatorname{rang}\left(\frac{\partial g}{\partial x}(x,y),\ \frac{\partial g}{\partial y}(x,y)\right)=$

$=\operatorname{rang}\begin{pmatrix}2x & 2y\end{pmatrix}=1,\ \forall\, x,y\in\mathbb{R}^2$ cu $x^2+y^2=1$ (nu poate fi 0)

(II) Definim funcția $L:\mathbb{R}^2\to\mathbb{R}$,

$L(x,y)=f(x,y)+\lambda g(x,y)$

FCT. LAGRANGE

$\Rightarrow L(x,y)=6-4x-3y+\lambda(x^2+y^2-1)$

(III) Rezolvăm sistemul:

$$\begin{cases}\frac{\partial L}{\partial x}(x,y)=0\\ \frac{\partial L}{\partial y}(x,y)=0\\ g(x,y)=0\end{cases}\Rightarrow\begin{cases}-4+\lambda 2x=0\\ -3+\lambda 2y=0\\ x^2+y^2=1\end{cases}$$

$\Rightarrow x = \frac{2}{\lambda}, y = \frac{3}{2\lambda}, \lambda \neq 0$

$x^2+y^2=1 \Rightarrow \frac{4}{\lambda^2} + \frac{9}{4\lambda^2} = 1 \Rightarrow$

$\Rightarrow 25 = 4\lambda^2 \Rightarrow \boxed{\lambda = \pm\frac{5}{2}}$

- $\lambda_1 = \frac{5}{2} \Rightarrow x = \frac{4}{5}, y = \frac{3}{5} \Rightarrow \left(\frac{4}{5}, \frac{3}{5}\right)$ este punct critic condiționat
- $\lambda_2 = -\frac{5}{2} \Rightarrow \left(-\frac{4}{5}, -\frac{3}{5}\right)$ punct critic condiționat.

(IV) $d^2L(x_i, y_i)$
- poz def $\Rightarrow (x_i, y_i)$ - MINIM LOCAL cond. cu leg. $g(x,y)=0$
- neg. def $\Rightarrow (x_i, y_i)$ - MAXIM LOCAL
- nedef $\Rightarrow (x_i, y_i)$ nu e punct de extrem

Deci:

$d^2L(x_i, y_i) = \frac{\partial^2 L}{\partial x^2}(x_i, y_i)dx^2 + \frac{\partial^2 L}{\partial y^2}(x_i, y_i)dy^2$

$+ 2 \cdot \frac{\partial^2 L}{\partial x \partial y}(x_i, y_i)dx\,dy$

- $\frac{\partial^2 L}{\partial x^2}(x, y) = 2\lambda$
- $\frac{\partial^2 L}{\partial y^2}(x, y) = 2\lambda$
- $\frac{\partial^2 L}{\partial x \partial y}(x, y) = 0$

$\Rightarrow$

$\Rightarrow d^2L(x_i, y_i) = 2\lambda dx^2 + 0 + 2\lambda dy^2 =$

$= 2\lambda dx^2 + 2\lambda dy^2$

- $d^2L\left(\frac{4}{5}, \frac{3}{5}\right) = 2 \cdot \frac{5}{2}(dx^2+dy^2) = 5(dx^2+dy^2) > 0$

$\Rightarrow \left(\frac{4}{5}, \frac{3}{5}\right)$ - MINIM LOCAL cu leg...

$\cdot\ d^2L\left(-\frac{4}{5},-\frac{3}{5}\right)=2\cdot\left(-\frac{5}{2}\right)(dx^2+dy^2)<0 \Rightarrow$

$\Rightarrow$ neg. def. $\Rightarrow$

$\Rightarrow \left(-\frac{4}{5},-\frac{3}{5}\right)$ punct de maxim local

$\boxed{2}$ Găsiți punctele de extrem local ale fct. $f:\mathbb{R}^3\to\mathbb{R}$, $f(x,y,z)=x-2y+2z$ cu legătura $x^2+y^2+z^2=9$.

Eu

3) Det. pct. de extrem ale funcției $f:\mathbb{R}^3 \to \mathbb{R}$, $f(x,y,z) = xyz$ cu legătura $x+y+z=1$, $x>0, y>0, z>0$

SOL:

Fie $D = \{(x,y,z) \in \mathbb{R}^3;\ x>0, y>0, z>0\}$

Fie $g: D \to \mathbb{R}$, $g(x,y,z) = x+y+z-1$

Evident $f, g \in \mathcal{C}^2(D)$ m. deschisă

I $\operatorname{rang}\left(\frac{\partial g}{\partial x}(x,y,z)\ \ \frac{\partial g}{\partial y}(x,y,z)\ \ \frac{\partial g}{\partial z}(x,y,z)\right)$

$= \operatorname{rang}(1\ \ 1\ \ 1) = 1,\ \forall (x,y,z) \in D$

II Def. fct. Lagrange $L: D \to \mathbb{R}$,

$L(x,y,z) = f(x,y,z) + \lambda g(x,y,z) =$

$= xyz + \lambda(x+y+z-1)$

III Rez. sistemul:

$$\begin{cases} \frac{\partial L}{\partial x} = 0 \\ \frac{\partial L}{\partial y} = 0 \\ \frac{\partial L}{\partial z} = 0 \\ g(x,y,z) = 0 \end{cases} \Rightarrow \begin{cases} yz + \lambda = 0 \Rightarrow \lambda = -yz \\ xz + \lambda = 0 \Rightarrow \lambda = -xz \\ xy + \lambda = 0 \Rightarrow \lambda = -xy \\ x+y+z = 1 \end{cases}$$

$yz = xz = xy \Rightarrow \underbrace{x}_{>0}(y-z) = 0 \Rightarrow \boxed{y=z}$

$\Rightarrow y = z = x \Rightarrow 3x = 1 \Rightarrow x = y = z = \frac{1}{3}$

$\Rightarrow \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$ punct critic condiționat

IV $d^2L\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right) = \frac{\partial^2 L}{\partial x^2}\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)dx^2 +$

$+ \frac{\partial^2 L}{\partial y^2}\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)dy^2 + \frac{\partial^2 L}{\partial z^2}\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)dz^2 +$

$+ 2 \cdot \frac{\partial^2 L}{\partial x \partial y}\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)dx\,dy + \ldots$

$\Rightarrow d^2L\left(\frac{1}{3},\frac{1}{3},\frac{1}{3}\right) = 2\cdot\frac{1}{3}dx\,dy + 2\cdot\frac{1}{3}dy\,dz$
$+\frac{2}{3}dx\,dz = \frac{2}{3}(dx\,dy + dy\,dz + dx\,dz) = ??$

Diferențiem legătura

Știm $g(x,y,z)=0 \Rightarrow dg(x,y,z)=0$,
$\forall (x,y,z)\in D \Rightarrow$

$\Rightarrow \frac{\partial g}{\partial x}dx + \frac{\partial g}{\partial y}dy + \frac{\partial g}{\partial z}dz = 0 \Rightarrow$

$\Rightarrow dx + dy + dz = 0 \Rightarrow$

$\Rightarrow dz = -dx - dy \Rightarrow$

$\Rightarrow d^2L\left(\frac{1}{3},\frac{1}{3},\frac{1}{3}\right) = \frac{2}{3}(\cancel{dx\,dy} - dy\,dx - dy^2 -$

Var. 1

$- dx^2 - \cancel{dx\,dy}) =$

$= -\frac{2}{3}(dx^2 + dy^2 + dy\,dx) =$

$= -\frac{2}{3}\Big(\underbrace{\left(dx + \frac{1}{2}dy\right)^2 + \left(\frac{\sqrt{3}}{2}dy\right)^2}_{>0}\Big) < 0$

$\Rightarrow$ neg. def $\Rightarrow$

$\Rightarrow \left(\frac{1}{3},\frac{1}{3},\frac{1}{3}\right)$ pct. max. local cu leg. $x+y+z=1$.

Var. 2

$d^2L\left(\frac{1}{3},\frac{1}{3},\frac{1}{3}\right) = -\frac{2}{3}(dx^2 + dx\,dy + dy^2) \Rightarrow$

$\Rightarrow H_{leg} = \begin{pmatrix} -\frac{2}{3} & -\frac{1}{3} \\ -\frac{1}{3} & -\frac{2}{3} \end{pmatrix} \quad \left.\begin{matrix}\Delta_1 < 0 \\ \Delta_2 > 0\end{matrix}\right| \Rightarrow$

$\Rightarrow \left(\frac{1}{3},\frac{1}{3},\frac{1}{3}\right)$ pct. max. local cu leg $x+y+z=1$

4. Fie $f: \mathbb{R}^3 \to \mathbb{R}$, $f(x,y,z) = xy + xz + yz$

Det. pct. de extrem local ale fct. cu leg. $-x+y+z=1$, $x-z=0$.

SOL:

Fie $g_1, g_2: \mathbb{R}^3 \to \mathbb{R}$,

$g_1(x,y,z) = -x+y+z-1$

$g_2(x,y,z) = x-z$

$g_1, g_2 \in \mathcal{C}^2(\underbrace{\mathbb{R}^3}_{\text{m. deschisă}})$

(I) $\text{rang}\begin{pmatrix} \frac{\partial g_1}{\partial x}(x,y,z) & \frac{\partial g_1}{\partial y} & \frac{\partial g_1}{\partial z} \\ \frac{\partial g_2}{\partial x} & \frac{\partial g_2}{\partial y} & \frac{\partial g_2}{\partial z} \end{pmatrix} =$

$= \text{rang}\begin{pmatrix} -1 & 1 & 1 \\ 1 & 0 & -1 \end{pmatrix} = 2, \forall x,y,z \in \mathbb{R}^3$

(II) Def. fct. Lagrange $L: \mathbb{R}^3 \to \mathbb{R}$,

$L(x,y,z) = f(x,y,z) + \lambda_1 g_1(x,y,z) +$

$+ \lambda_2 g_2(x,y,z) =$

$= xy + yz + zx + \lambda_1(-x+y+z-1) +$

$+ \lambda_2(x-z)$

(III) Rez. sistemul:

$$\begin{cases} \frac{\partial L}{\partial x} = 0 \\ \frac{\partial L}{\partial y} = 0 \\ \frac{\partial L}{\partial z} = 0 \\ g_1(x,y,z) = 0 \\ g_2(x,y,z) = 0 \end{cases} \Leftrightarrow \begin{cases} y+z-\lambda_1+\lambda_2=0 \\ x+z+\lambda_1=0 \\ x+y+\lambda_1-\lambda_2=0 \\ -x+y+z-1=0 \\ x-z=0 \Rightarrow x=z \end{cases}$$

$$\Rightarrow \begin{cases} y+x-\lambda_1+\lambda_2=0 \\ 2x+\lambda_1=0 \\ x+y+\lambda_1-\lambda_2=0 \\ y=1 \end{cases}$$

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$$\begin{cases} \lambda + x - \ldots \\ 2x + \lambda_1 = 0 \\ x + 1 + \lambda_1 - \lambda_2 = 0 \end{cases} \Rightarrow \boxed{x = -1} = z. \quad \Rightarrow \boxed{\begin{matrix}\lambda_1 = -2 \\ \lambda_2 = -2\end{matrix}}$$

$\Rightarrow (-1, 1, -1)$ pct critic cond.

$d^2L(-1,1,-1) = 2(dx\,dy + dx\,dz + dy\,dz)$

(IV)

(V) Diferențiem legăturile:

- $g_1(x,y,z) = 0 \Rightarrow$

$\Rightarrow dg_1(x,y,z) = 0 \Rightarrow$

$\Rightarrow \frac{\partial g_1}{\partial x}(x,y,z)dx + \frac{\partial g_1}{\partial y}(x,y,z)dy + \frac{\partial g_1}{\partial z}(x,y,z)dz = 0 \Rightarrow$

$\Rightarrow -dx + dy + dz = 0 \Rightarrow \boxed{dz = dx - dy}$

- $g_2(x,y,z) = 0 \Rightarrow dg_2(x,y,z) = 0 \Rightarrow$

$\Rightarrow dx - dz = 0 \Rightarrow dy = 0 \Rightarrow dx = dz$

$\Rightarrow d^2L(-1,1,-1) = 2 \cdot dx^2 > 0 \Rightarrow$

$\Rightarrow (-1,1,-1)$ - punct de minim local pentru f cu legăturile $g_1 = 0, g_2 = 0$.

5. Fie $f: \mathbb{R}^2 \to \mathbb{R}$, $f(x,y) = x^2 - y^2$. Det. val. extreme ale lui f pe mulțimea $K = \{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2 \le 4\}$

$\left(\sup_{(x,y)\in K} f(x,y) = ?, \inf_{(x,y)\in K} f(x,y) = ? \right)$

Sol:

f continuă, K - compactă $\Rightarrow$

$\Rightarrow f$ - mărg. și își atinge marginile $\Rightarrow \sup_{(x,y)\in K} f(x,y) = \max_{(x,y)\in K} f(x,y)$

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① Punctele de extrem din interior
Fie K: $\overset{\circ}{K}=\{(x,y)\in\mathbb{R}^2 \mid x^2+y^2<4\}$

$$\begin{cases} \frac{\partial f}{\partial x}(x,y)=0 \\ \frac{\partial f}{\partial y}(x,y)=0 \end{cases} \Rightarrow \begin{cases} 2x=0 \\ -2y=0 \end{cases} \Rightarrow$$

$\Rightarrow (0,0)\in\overset{\circ}{K}$ punct critic

$$H(x,y)=\begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix} \Rightarrow$$

$$\Rightarrow H(0,0)=\begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix} \quad \begin{matrix} \Delta_1=2>0 \\ \Delta_2=-4<0 \end{matrix} \Bigg\} \Rightarrow$$

$\Rightarrow (0,0)$ nu e punct de extrem

② Punctele de extrem situate pe
$Fr K=\{(x,y)\in\mathbb{R}^2 \mid x^2+y^2=4\}$
legătura

Fie $g:\mathbb{R}^2\to\mathbb{R}$, $g(x,y)=x^2+y^2-4$
$f,g\in\mathcal{C}^2(\mathbb{R}^2)$
c.m.d.

- $\text{rang}\begin{pmatrix} \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} \end{pmatrix}=\text{rang}\begin{pmatrix} 2x & 2y \end{pmatrix}=$
$=1, \forall (x,y)\in Fr K$
- Def. fct. Lagrange $L:\mathbb{R}^2\to\mathbb{R}$
$L(x,y)=f(x,y)+\lambda g(x,y)=$
$=x^2-y^2+\lambda(x^2+y^2-4)$
- Rez sistemul:

$$\begin{cases} \frac{\partial L}{\partial x}=0 \\ \frac{\partial L}{\partial y}=0 \\ g(x,y)=0 \end{cases} \Rightarrow \begin{cases} 2x+2\lambda x=0 \\ -2y+2\lambda y=0 \\ x^2+y^2-4=0 \end{cases}$$

$x(1+\lambda)=0 \Rightarrow x=0$ sau $\lambda=-1$
$y(-1+\lambda)=0$

• $x=0 \Rightarrow$ ...
$y=\pm 2 \Rightarrow \boxed{\lambda=1} \Rightarrow (0,2),(0,-2)$
pct. critice cond.

• $\lambda=-1 \Rightarrow y=0 \Rightarrow x^2=4 \Rightarrow x=\pm 2 \Rightarrow$
$\Rightarrow (2,0),(-2,0)$ pct. critice cond.

• $d^2L(x_i,y_i) = \frac{\partial^2 L}{\partial x^2}(x_i,y_i)dx^2 + 2\frac{\partial^2 L}{\partial x \partial y}(x_i,y_i)$
$d_x dy + \frac{\partial^2 L}{\partial y^2}(x_i,y_i)dy^2 =$
$= (2+2\lambda)dx^2 + (-2+2\lambda)dy^2 + 0\,dx\,dy$
$= (2+2\lambda)dx^2 + (-2+2\lambda)dy^2$.

$d^2L(0,2) \overset{\lambda=1}{=} 4dx^2 > 0 \Rightarrow (0,2)$ pct. de minim local
$d^2L(0,-2) \overset{\lambda=1}{=} 4dx^2 > 0 \Rightarrow (0,-2)$ pct de minim local
$d^2L(2,0) = d^2L(-2,0) \overset{\lambda=-1}{=} -4dy^2 < 0 \Rightarrow$
$\Rightarrow \underbrace{(2,0),(-2,0)}_{\text{pct. FxK}}$ - pct de maxim local

$f(0,-2) = -4$ $\quad f(2,0) = 4$
$f(0,2) = -4$ $\quad f(-2,0) = 4$

$\Rightarrow \sup_{(x,y)\in K} f(x,y) = \max_{(x,y)\in K} f(x,y) = 4$
$\inf_{(x,y)\in K} f(x,y) = \min_{(x,y)\in K} f(x,y) = -4$.

6. Extremele funcției $f:\mathbb{R}^2 \to \mathbb{R}$, $f(x,y)=xy$ pe mulțimea $K=\{(x,y)\in\mathbb{R}^2 \mid x^2+y^2\le 1\}$

Det. $\sup_{(x,y)\in K} f(x,y)$, $\inf_{(x,y)\in K} f(x,y)$

SOL:

f - continuă, K compactă $\Rightarrow$ f - mărginită și își atinge marginile $\Rightarrow$

$\Rightarrow \sup_{(x,y)\in K} f(x,y) = \max_{(x,y)\in K} f(x,y)$

$\inf_{(x,y)\in K} f(x,y) = \min_{(x,y)\in K} f(x,y)$

(I) Punctele de extrem din interiorul lui K: $\overset{\circ}{K}=\{(x,y)\in\mathbb{R}^2 \mid x^2+y^2<1\}$

$$\begin{cases} \frac{\partial f}{\partial x}(x,y)=0 \\ \frac{\partial f}{\partial y}(x,y)=0 \end{cases} \Rightarrow \begin{cases} y=0 \\ x=0 \end{cases} \Rightarrow$$

$\Rightarrow (0,0)\in\overset{\circ}{K}$ punct critic

$$H(x,y)=\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \Rightarrow$$

$$\Rightarrow H(0,0)=\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \Rightarrow \left.\begin{array}{l} \Delta_1=0 \\ \Delta_2=-1<0 \end{array}\right| \Rightarrow$$

$\Rightarrow$ pct. șa

(II) Punctele de extrem ale lui f pe $\operatorname{Fr} K=\{(x,y)\in\mathbb{R}^2 \mid x^2+y^2=1\}$

Fie $g:\mathbb{R}^2\to\mathbb{R}$, $g(x,y)=x^2+y^2-1$

$f,g\in C^2(\mathbb{R}^2)$ ș.m.d.

• rang $\left(\frac{\partial g}{\partial x} \quad \frac{\partial g}{\partial y}\right)$ = rang $(2x \quad 2y)$ $=1$, $\forall (x,y)\in\mathbb{R}^2$

$L(x,y) = f(x,y) + \lambda g(x,y)$; $L: \mathbb{R}^2 \to \mathbb{R}$.
$= xy + \lambda(x^2+y^2-1)$

• Rez. sistemul:

$$\begin{cases} \frac{\partial L}{\partial x} = 0 \\ \frac{\partial L}{\partial y} = 0 \\ g(x,y) = 0 \end{cases} \Rightarrow \begin{cases} y + 2\lambda x = 0 \\ x + 2\lambda y = 0 \Rightarrow y = -2\lambda x \\ x^2+y^2-1=0 \end{cases}$$

$\Rightarrow x - 4\lambda^2 x = 0 \Rightarrow x(1-4\lambda^2) = 0 \Rightarrow$

$\Rightarrow x = 0$ sau $1 = 4\lambda^2$, adică $\lambda = \pm\frac{1}{2}$.

• $x=0 \Rightarrow y = 0 \Rightarrow x^2+y^2 = 0$ $\not\!\exists$

• $\boxed{\lambda = \frac{1}{2}} \Rightarrow y = -x \Rightarrow 2x^2 = 1 \Rightarrow x = \pm\frac{1}{\sqrt{2}}$

$\Rightarrow \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right), \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ pct. critice cond.

• $\boxed{\lambda = -\frac{1}{2}} \Rightarrow y = x \Rightarrow 2x^2 = 1 \Rightarrow x = \pm\frac{1}{\sqrt{2}}$

$\Rightarrow \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right), \left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$ pct. critice cond

• $d^2L(x_i,y_i) = \frac{\partial^2 L}{\partial x^2}(x_i,y_i)\,dx^2 + 2\cdot\frac{\partial^2 L}{\partial x \partial y}(x_i,y_i)\,dx\,dy + \frac{\partial^2 L}{\partial y^2}(x_i,y_i)\,dy^2$

$= 2\lambda\, dx^2 + 2\cdot 1\, dx\,dy + 2\lambda\, dy^2 =$

$= 2(\lambda\, dx^2 + dx\,dy + \lambda\, dy^2)$

$d^2L\left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right) = d^2L\left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) \overset{\lambda=\frac{1}{2}}{=}$

$= 2\left(\frac{1}{2}dx^2 + dx\,dy + \frac{1}{2}dy^2\right) =$

$= dx^2 + 2dx\,dy + dy^2 = (dx+dy)^2 \geq 0$

$\Rightarrow \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right), \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ - pct de minim local ale lui f pe $\mathbb{R} \times \mathbb{R}$

$d^2L\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) = d^2L\left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right) \overset{\lambda=-\frac{1}{2}}{=}$

$= -dx^2 + 2dx\,dy - dy^2 = -(dx-dy)^2 \leq 0$

$\Rightarrow \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right), \left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$ - pct de max local ale lui f pe $\mathbb{R} \times \mathbb{R}$

$f\left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right) = -\frac{1}{2}$ $f\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) = \frac{1}{2}$

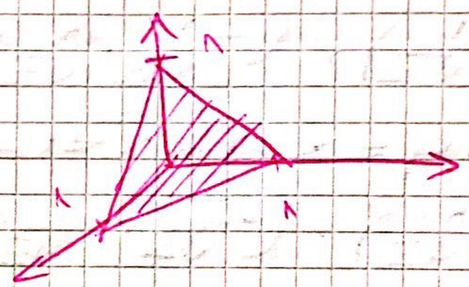
$f\left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) = -\frac{1}{2}$ $f\left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right) = \frac{1}{2}$

$\Rightarrow \sup\limits_{(x,y)\in K} f(x,y) = \max\limits_{(x,y)\in K} f(x,y) = \frac{1}{2}$

$\inf\limits_{(x,y)\in K} f(x,y) = \min\limits_{(x,y)\in K} f(x,y) = -\frac{1}{2}$

7. Extremele funcției $f(x,y,z) = x^2+y^2+z^2 + 2x + 4y - 6z$ pe mulț. $K = \{(x,y,z) \mid x + y + z \leq 1\}$

SOL:



Pe $\overset{\circ}{K} = \{(x,y,z) \mid x+y+z<1\}$

$$\begin{cases} \frac{\partial f}{\partial x}(x,y,z) = 0 \\ \frac{\partial f}{\partial y}(x,y,z) = 0 \\ \frac{\partial f}{\partial z}(x,y,z) = 0 \end{cases} \Rightarrow \begin{cases} 2x+2=0 \\ 2y+4=0 \\ 2z-6=0 \end{cases} \Rightarrow \begin{cases} x=-1 \\ y=-2 \\ z=3 \end{cases}$$

$\Rightarrow (-1,-2,3)$ pct critic

$$H(x,y,z) = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} \Rightarrow$$

$$\Rightarrow H(-1,-2,3) = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} \Rightarrow \begin{matrix} \Delta 1 = 2 > 0 \\ \Delta 2 = 4 > 0 \\ \Delta 3 = 8 > 0 \end{matrix} \Rightarrow$$

$\Rightarrow (-1,-2,3)$ pct de minim local pt f pe $\overset{\circ}{K}$

• Pe $A = \{(x,y,z) \mid x+y+z=1\}$

Fie $g: \mathbb{R}^3 \to \mathbb{R}$, $g(x,y,z) = x+y+z-1$

$f, g \in \mathcal{C}^2(\mathbb{R}^3)$ m. deschisă

$\text{rang}\left(\frac{\partial g}{\partial x}(x,y,z) \quad \frac{\partial g}{\partial y}(x,y,z) \quad \frac{\partial g}{\partial z}(x,y,z) \right)$

$= \text{rang}\begin{pmatrix} 1 & 1 & 1 \end{pmatrix} = 1, \forall (x,y,z) \in \mathbb{R}^3$

• Def fct. Lagrange $L: \mathbb{R}^3 \to \mathbb{R}$

$L(x,y,z) = f(x,y,z) + \lambda g(x,y,z) =$

$= x^2+y^2+z^2+2x+4y-6z +$

$+ \lambda(x+y+z-1)$

• Rez sist.:

$$\begin{cases} \frac{\partial L}{\partial x}(x,y,z) = 0 \\ \frac{\partial L}{\partial y} = 0 \\ \frac{\partial L}{\partial z} = 0 \\ g(x,y,z) = 0 \end{cases} \Rightarrow \begin{cases} 2x+2+\lambda = 0 \\ 2y+4+\lambda = 0 \\ 2z-6+\lambda = 0. \\ x+y+z-1 = 0 \ (*) \end{cases}$$

$x = -1 - \frac{\lambda}{2}, \ y = -2 - \frac{\lambda}{2}, \ z = 3 - \frac{\lambda}{2}$

$(*) \Rightarrow \cancel{-1} - \frac{\lambda}{2} \cancel{-2} - \frac{\lambda}{2} + \cancel{3} - \frac{\lambda}{2} - 1 = 0 \Rightarrow$

$\Rightarrow 3\frac{\lambda}{2} = -1 \Rightarrow \boxed{\lambda = -\frac{2}{3}} \Rightarrow$

$\Rightarrow x = -1 + \frac{1}{3} = -\frac{2}{3}$

$y = -2 + \frac{1}{3} = -\frac{5}{3}$

$z = 3 + \frac{1}{3} = \frac{10}{3}$

$\Rightarrow \left(-\frac{2}{3}, -\frac{5}{3}, \frac{10}{3}\right)$ pct. critic

• $d^2L\left(-\frac{2}{3}, -\frac{5}{3}, \frac{10}{3}\right) = \frac{\partial^2 L}{\partial x^2}\left(-\frac{2}{3}, -\frac{5}{3}, \frac{10}{3}\right) dx^2 +$

$+ \frac{\partial^2 L}{\partial y^2}\left(-\frac{2}{3}, -\frac{5}{3}, \frac{10}{3}\right) dy^2 + \frac{\partial^2 L}{\partial z^2}\left(-\frac{2}{3}, -\frac{5}{3}, \frac{10}{3}\right) dz^2 +$

$+ 2\frac{\partial^2 L}{\partial x \partial y}\left(-\frac{2}{3}, -\frac{5}{3}, \frac{10}{3}\right) dx dy + 2 \cdot \frac{\partial^2 L}{\partial y \partial z}(..) dy dz$

$+ 2 \cdot \frac{\partial^2 L}{\partial x \partial z}(..) dx dz =$

$= 2dx^2 + 2dy^2 + 2dz^2 + 2 \cdot 0 dx dy + 2 \cdot 0 dy dz$

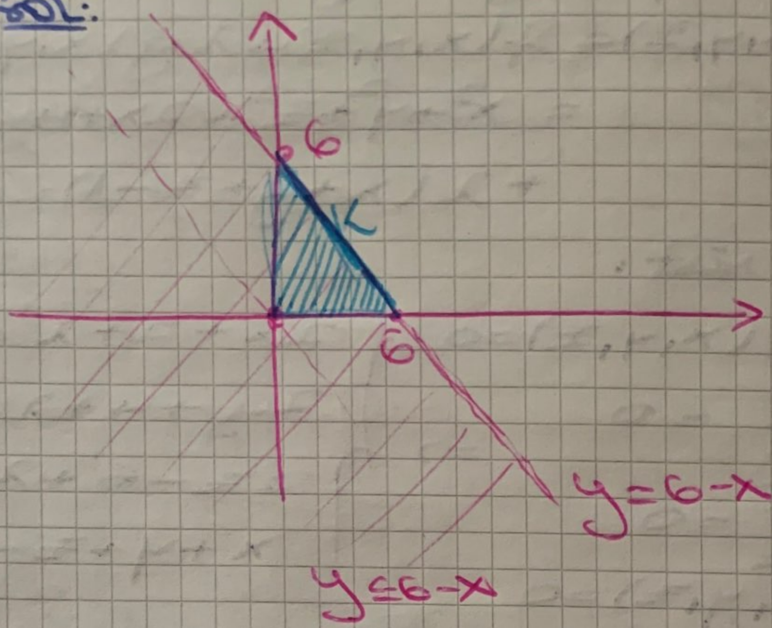
$+ 2 \cdot 0 \cdot dx dz =$

$\Rightarrow d^2L\left(-\frac{2}{3}, -\frac{5}{3}, \frac{10}{3}\right) = 2(dx^2+dy^2+dz^2)$
$>0 \Rightarrow \left(-\frac{2}{3}, -\frac{5}{3}, \frac{10}{3}\right)$ pct de minim local pe $T_{2}L$

8. Extremele funcției $f(x,y) = 4x - x^2 + 6y - y^2$ pe $K = \{(x,y) \in \mathbb{R}^2 \mid x \geq 0, y \geq 0, x + y \leq 6\}$

$y \leq 6 - x$

<u>SOL</u>:



① $\overset{\circ}{K} = \{(x,y) \mid x > 0, y > 0, x+y < 6\}$

$$\begin{cases} \frac{\partial f}{\partial x}(x,y) = 0 \\ \frac{\partial f}{\partial y}(x,y) = 0 \end{cases} \Rightarrow \begin{cases} 4 - 2x = 0 \Rightarrow x = 2 \\ 6 - 2y = 0 \Rightarrow y = 3 \end{cases}$$

$\Rightarrow (2,3) \in \overset{\circ}{K}$ pct. critic

$H(x,y) = \begin{pmatrix} -2 & 0 \\ 0 & -2 \end{pmatrix} \Rightarrow$

$\Rightarrow H(2,3) = \begin{pmatrix} -2 & 0 \\ 0 & -2 \end{pmatrix} \Rightarrow \begin{rcases} \Delta_1 = -2 < 0 \\ \Delta_2 = 4 > 0 \end{rcases} \Rightarrow$

$\Rightarrow (2,3)$ pct de maxim local

11) Fr K $= \{(x,y) \mid x+y=6\} \cup \{(x,0) \mid x\in[0,6]\}$
$x\in[0,6]$ $\cup \{(0,y) \mid y\in[0,6]\}$

$\cdot \{(x,y)\in\mathbb{R}^2 \mid x+y=6, x\in[0,6]\}$

A_1: Fie $g: A_1 \to \mathbb{R}$, $g(x,y)=x+y-6$

$f, g \in \mathcal{C}^2(A_1)$
$\hookrightarrow$ m.d.

- $\operatorname{rang}\left(\frac{\partial g}{\partial x} \ \ \frac{\partial g}{\partial y}\right) = \operatorname{rang}(1 \ \ 1) = 1$,
  $\forall (x,y)\in A_1$

- Fie $L: A_1 \to \mathbb{R}$, $L(x,y) = f(x,y) +$
  $+ \lambda g(x,y) = 4x - x^2 + 6y - y^2 + \lambda(x+y-6)$

$$\begin{cases} \frac{\partial L}{\partial x} = 0 \\ \frac{\partial L}{\partial y} = 0 \\ g(x,y) = 0 \end{cases} \Rightarrow \begin{cases} 4-2x+\lambda = 0 \\ 6-2y+\lambda = 0 \\ x+y=6 \end{cases} \begin{matrix} \Rightarrow x = 2+\frac{\lambda}{2} \\ y = 3+\frac{\lambda}{2} \\ \Rightarrow 5+\lambda = 6 \Rightarrow \\ \Rightarrow \lambda = 1 \end{matrix}$$

$\Rightarrow \left(\frac{5}{2}, \frac{7}{2}\right)$ - pct. critic

$$d^2L\left(\frac{5}{2},\frac{7}{2}\right) = \frac{\partial^2 L}{\partial x^2}\left(\frac{5}{2},\frac{7}{2}\right)dx^2 + \frac{\partial^2 L}{\partial y^2}\left(\frac{5}{2},\frac{7}{2}\right)dy^2 +$$
$$+ 2\cdot\frac{\partial^2 L}{\partial x\partial y}\left(\frac{5}{2},\frac{7}{2}\right)dx\,dy =$$
$$= -2dx^2 - 2dy^2 = -2(dx^2+dy^2) < 0$$

$\Rightarrow \left(\frac{5}{2},\frac{7}{2}\right)$ - pct de maxim pe Fr K

- $A_2 = \{(x,0)\in\mathbb{R}^2 \mid x\in[0,6]\}$
  $f(x,y) = f(x,0) = 4x - x^2 = h_1(x)$
  $x\in[0,6]$

$h_1'(x) = 4-2x = 0$
$\Rightarrow x = 2$

x	0		2		6
h'	$+$	$+$	$+\ 0$	$-\ -\ -$	
h	0	$\nearrow$	4	$\searrow$	-12

pct $(2,0)$ este pct. de maxim pt. f pe A_2

• $A_3 = \{(0,y) \mid y \in [0,6]\}$

$f(x,y) = f(0,y) = 6y - y^2 = h_2(y)$

$h_2'(y) = 6 - 2y = 0 \Rightarrow y = 3 \quad y \in [0,6]$

y	0			3		6
h_2'	+	+	+	0	−	−
h_2		$\nearrow$		9	$\searrow$	

$\Rightarrow$ $(0,3)$ punct de maxim pt. f pe A_3.

$\Rightarrow$ Punctele de extrem pe K sunt:

$(2,3), \left(\frac{5}{2}, \frac{7}{2}\right), (0,3), (2,0)$.

EX. CALCUL DIF.

1. i) $f(x,y) = y\,\varphi(x^2 - y^2)$

$$\frac{1}{x} \cdot \frac{\partial f}{\partial x} + \frac{1}{y} \cdot \frac{\partial f}{\partial y} = \frac{1}{y^2} \cdot f$$

Fie $u : \mathbb{R}^2 \to \mathbb{R}$, $u(x,y) = x^2 - y^2$

$$\frac{\partial f}{\partial x} = y \cdot \frac{\partial \varphi}{\partial u} \cdot \frac{\partial u}{\partial x} =$$

$$= y \cdot \frac{\partial \varphi}{\partial u} \cdot 2x$$

$$\frac{\partial f}{\partial y} = \varphi(u(x,y)) + y \cdot \frac{\partial \varphi}{\partial u}(-2y)$$

$$\Rightarrow \frac{1}{x} \cdot \frac{\partial f}{\partial x} + \frac{1}{y} \cdot \frac{\partial f}{\partial y} =$$

$$= \frac{1}{x}\left(y \cdot \frac{\partial \varphi}{\partial u} \cdot 2x\right) + \frac{1}{y}\left(\varphi(x^2 - y^2) - 2y^2 \cdot \frac{\partial \varphi}{\partial u}\right)$$